

EETS 7302 TCP/IP Network Administration

Session 5 Packet Delivery

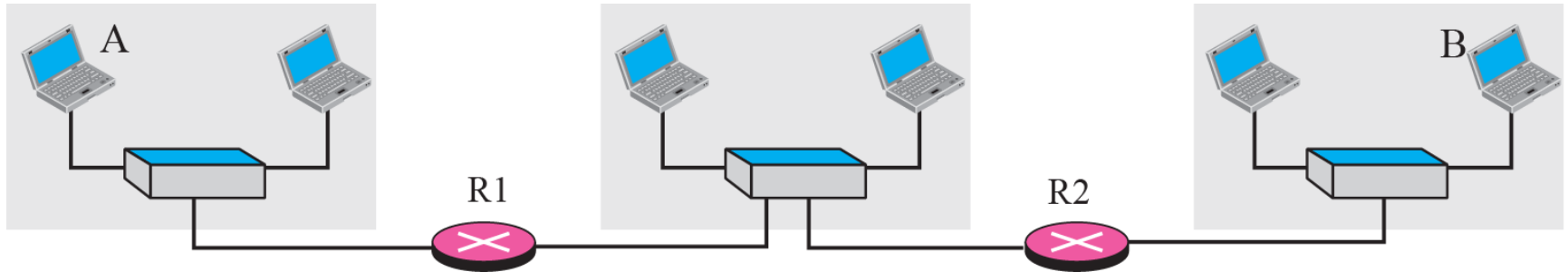
PACKET DELIVERY

- **The network layer supervises the handling of the packets by the underlying physical networks.**

FORWARDING

- **Forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device).**
- **Although the IP protocol was originally designed as a connectionless protocol, today the tendency is to use IP as a connection-oriented protocol.**

Next-hop method



A

Destination	Route
Host B	R1, R2, Host B

R1

Destination	Route
Host B	R2, Host B

R2

Destination	Route
Host B	Host B

a. Routing tables based on route

A

Destination	Next Hop
Host B	R1

R1

Destination	Next Hop
Host B	R2

R2

Destination	Next Hop
Host B	---

b. Routing tables based on next hop

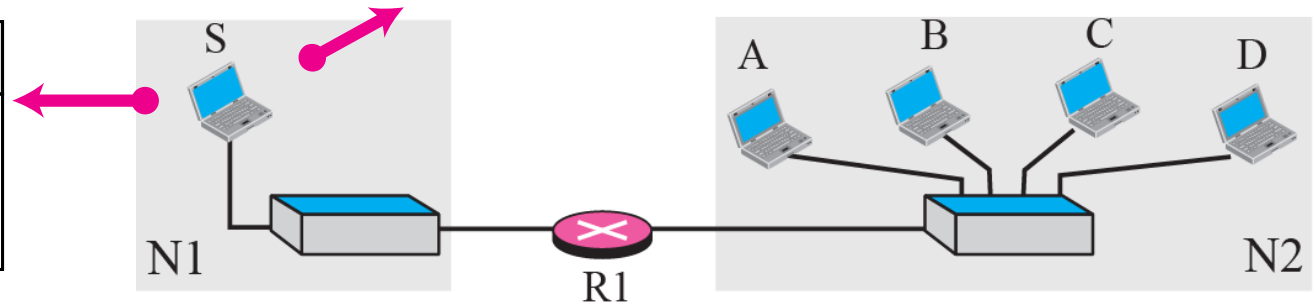
Network-specific method

Host-specific routing table for host S

Destination	Next Hop
A	R1
B	R1
C	R1
D	R1

Network-specific routing table for host S

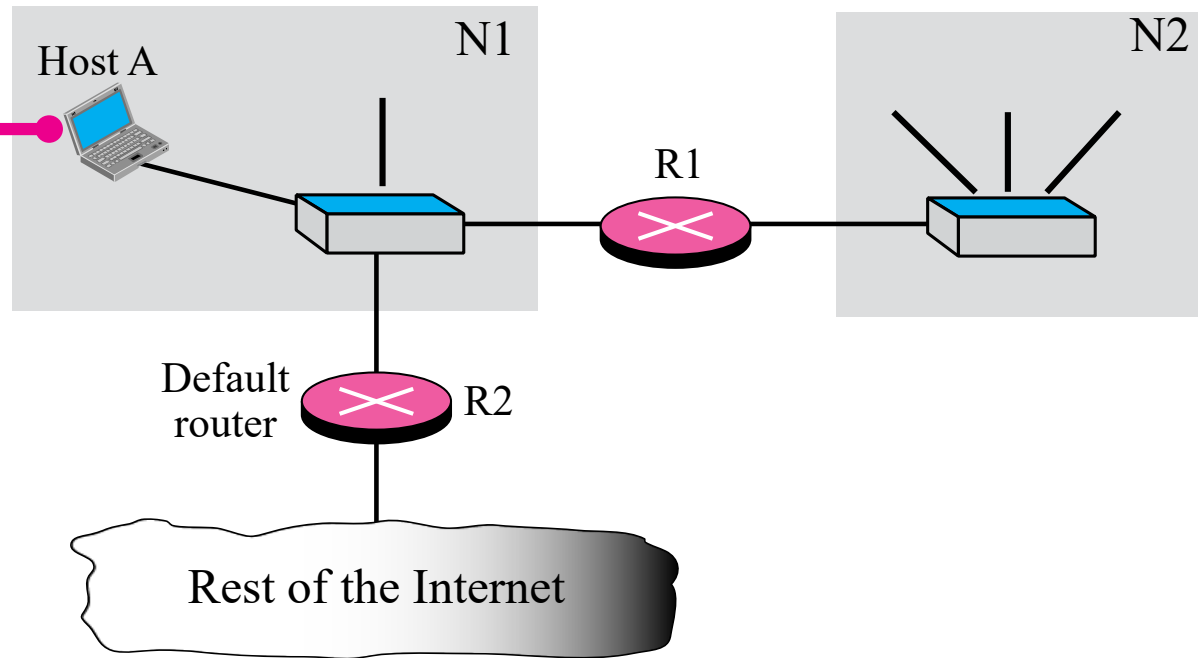
Destination	Next Hop
N2	R1



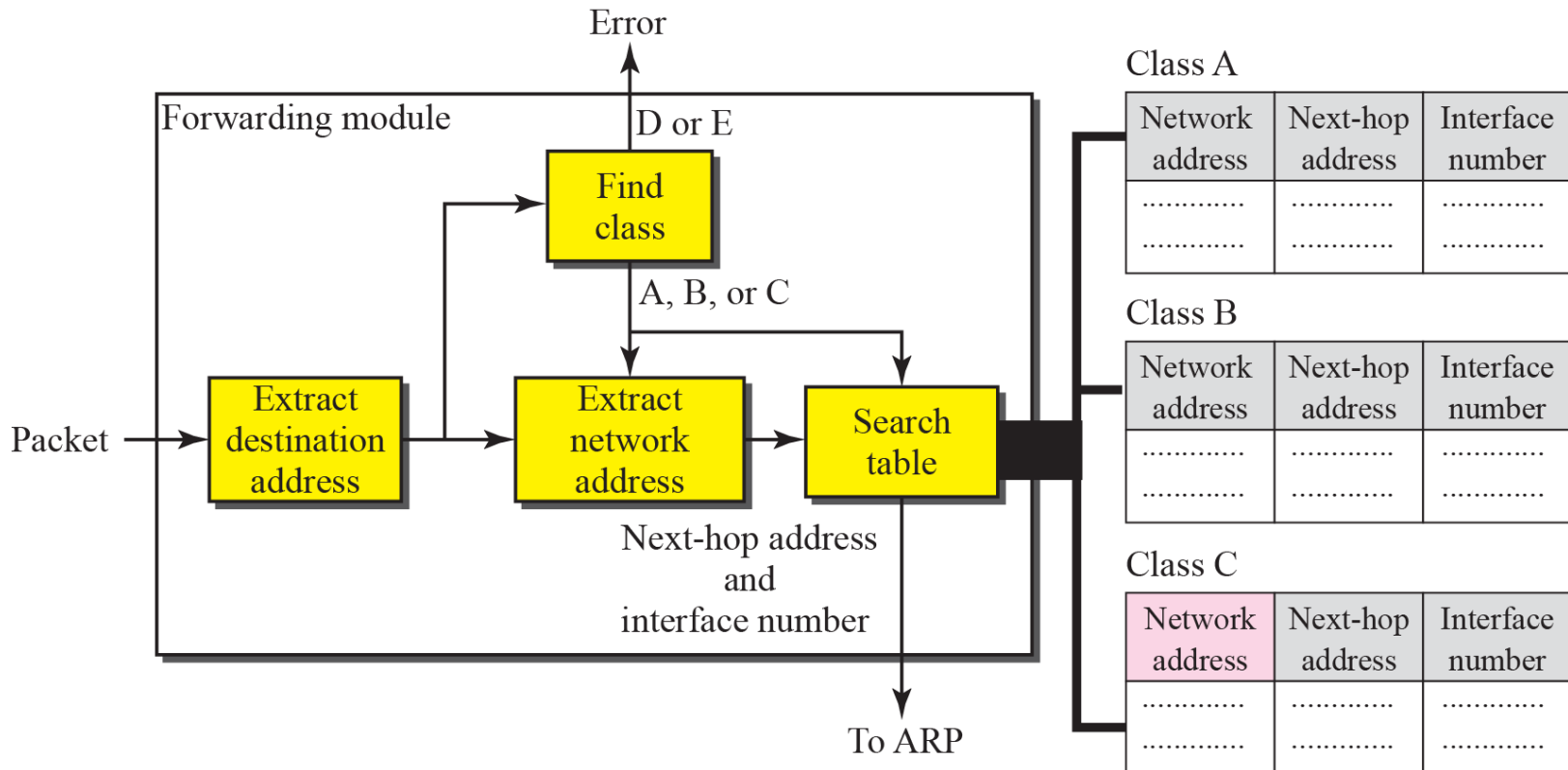
Default routing

Routing table for host A

Destination	Next Hop
N2	R1
.....	
Default	R2

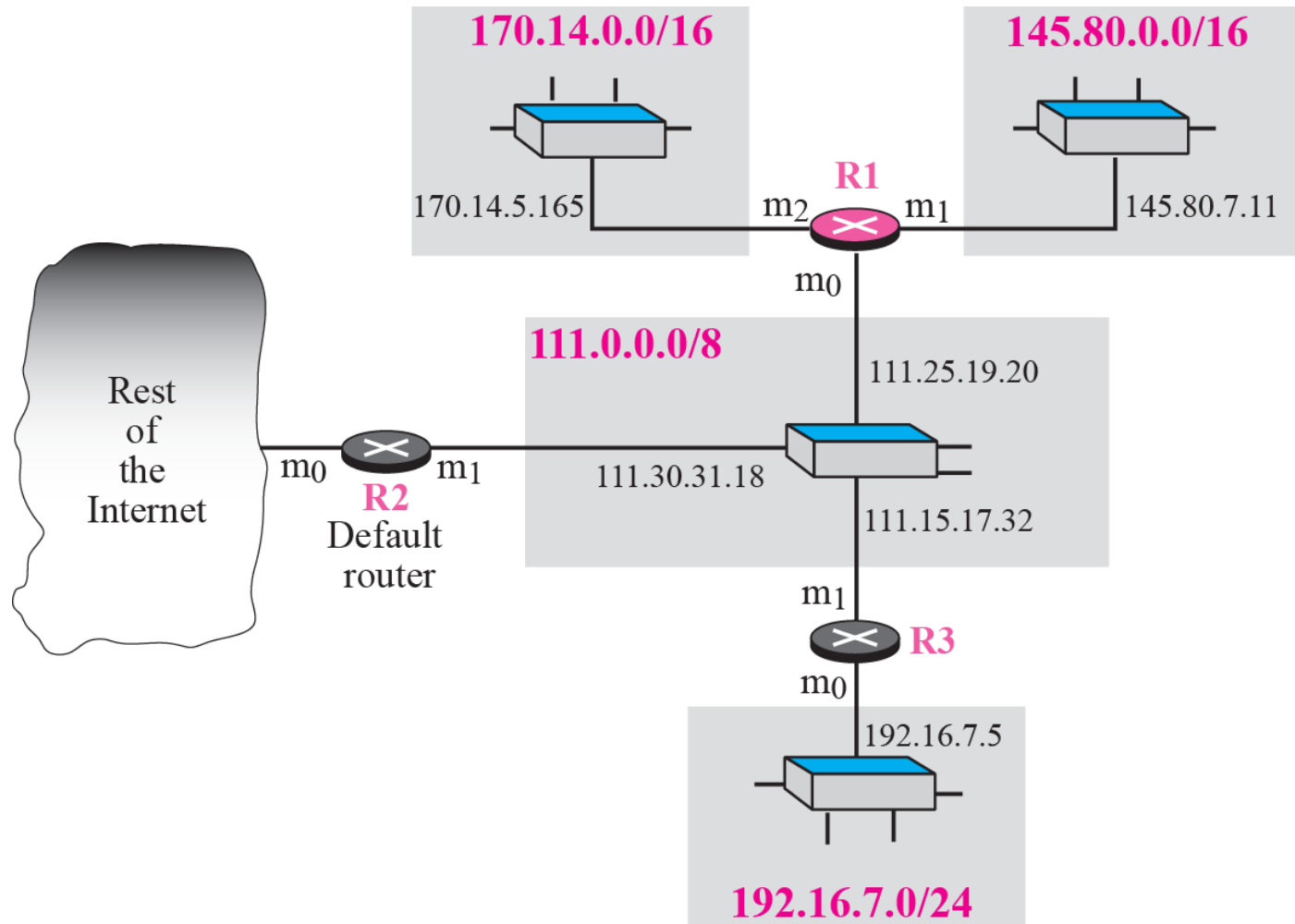


Simplified forwarding algorithm in classful address without subnetting



Example: 5min. exercise

Network below shows an imaginary part of the Internet.
Show the routing tables for router R1.



Example: 5min. exercise

Solution

three tables used by router R1. Note that some entries in the next-hop address column are empty because in these cases, the destination is in the same network to which the router is connected (direct delivery). In these cases, the next-hop address used by ARP is simply the destination address of the packet

Class A

Network address	Next-hop address	Interface
111.0.0.0	-----	m0

Class C

Network address	Next-hop address	Interface
192.16.7.0	111.15.17.32	m0

Class B

Network address	Next-hop address	Interface
145.80.0.0	-----	m1
170.14.0.0	-----	m2

Default: 111.30.31.18, m0

Example

Router R1 receives a packet with destination address 192.16.7.14. Show how the packet is forwarded.

Solution

The destination address is

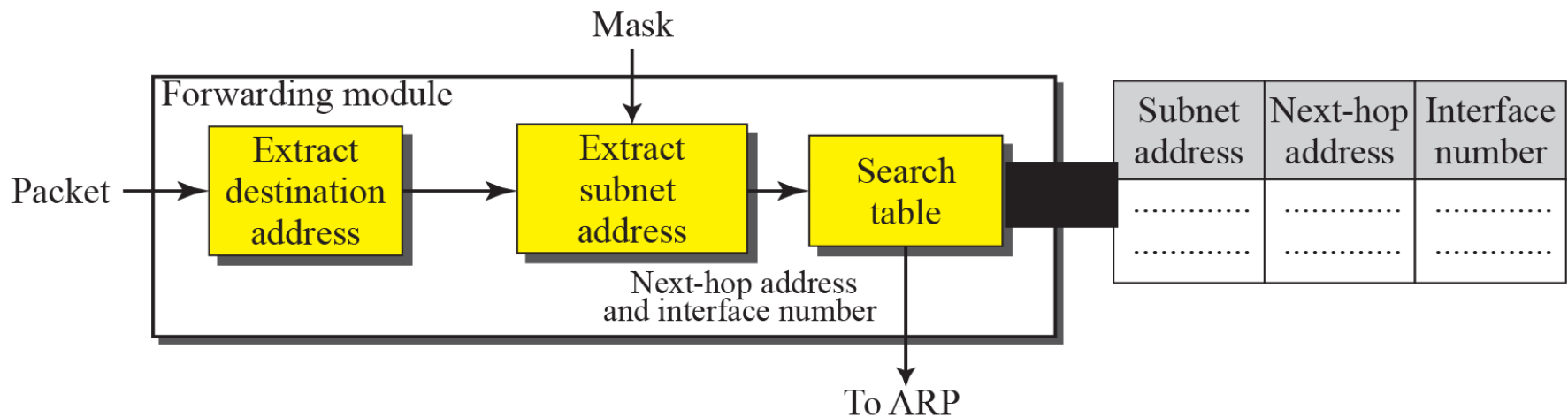
11000000 00010000 00000111 00001110.

A copy of the address is shifted 28 bits to the right. The result is

00000000 00000000 00000000 00001100 or **12**.

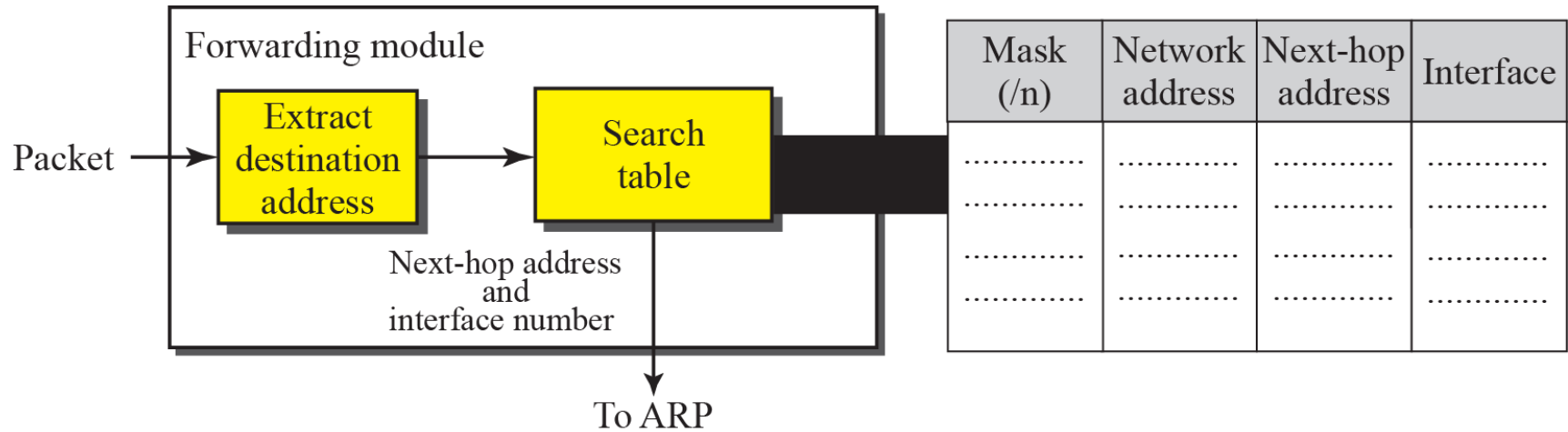
The destination network is class C. The network address is extracted by masking off the leftmost 24 bits of the destination address; the result is 192.16.7.0. The table for Class C is searched. The network address is found in the first row. The next-hop address 111.15.17.32. and the interface m0 are passed to ARP

Simplified forwarding algorithm in classful address with subnetting



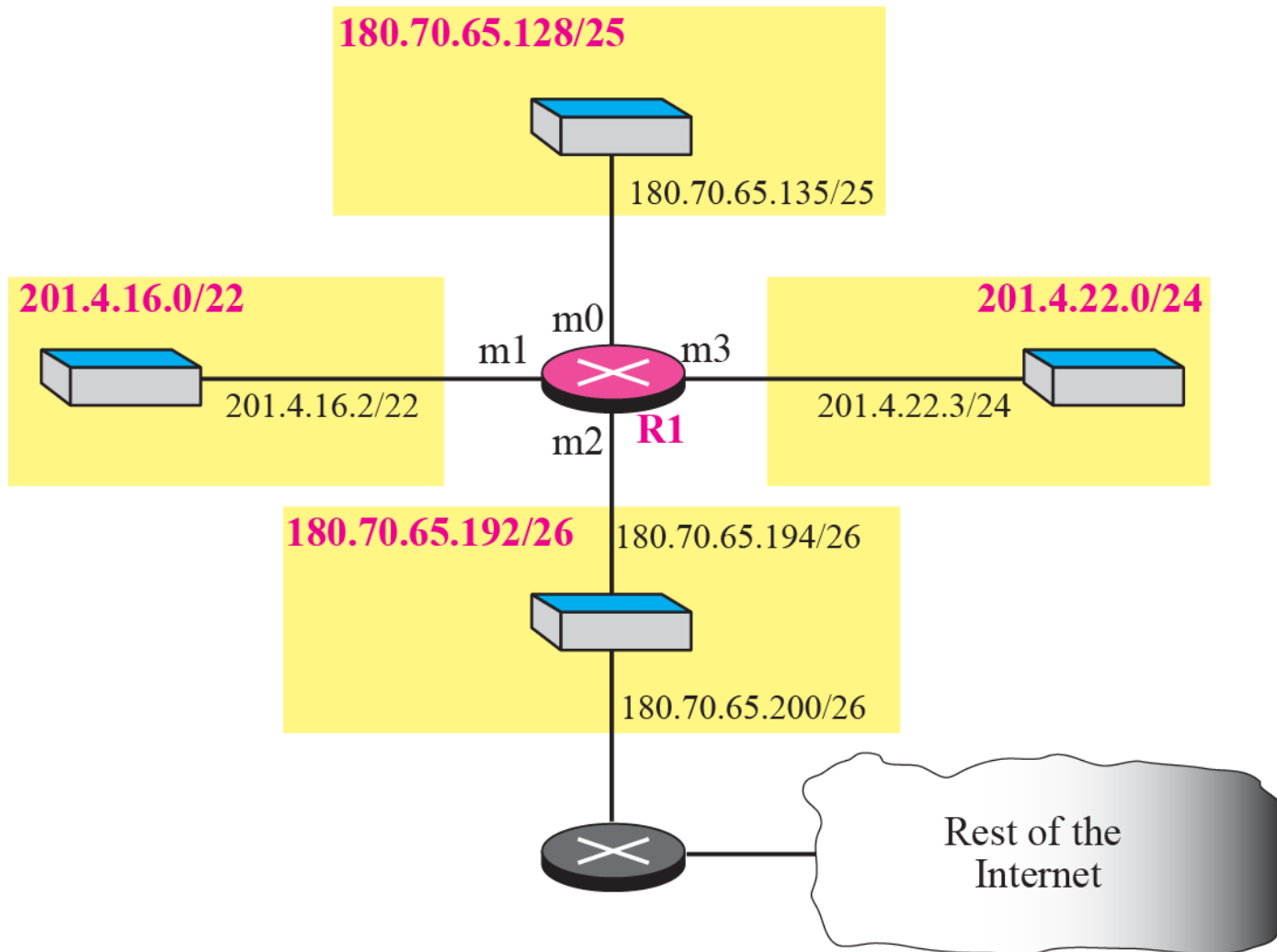
In classful addressing we can have a routing table with three columns; in classless addressing, We need at least four columns.

Simplified forwarding algorithm in classless address



Example

Make a routing table for router R1 in the below network



Example

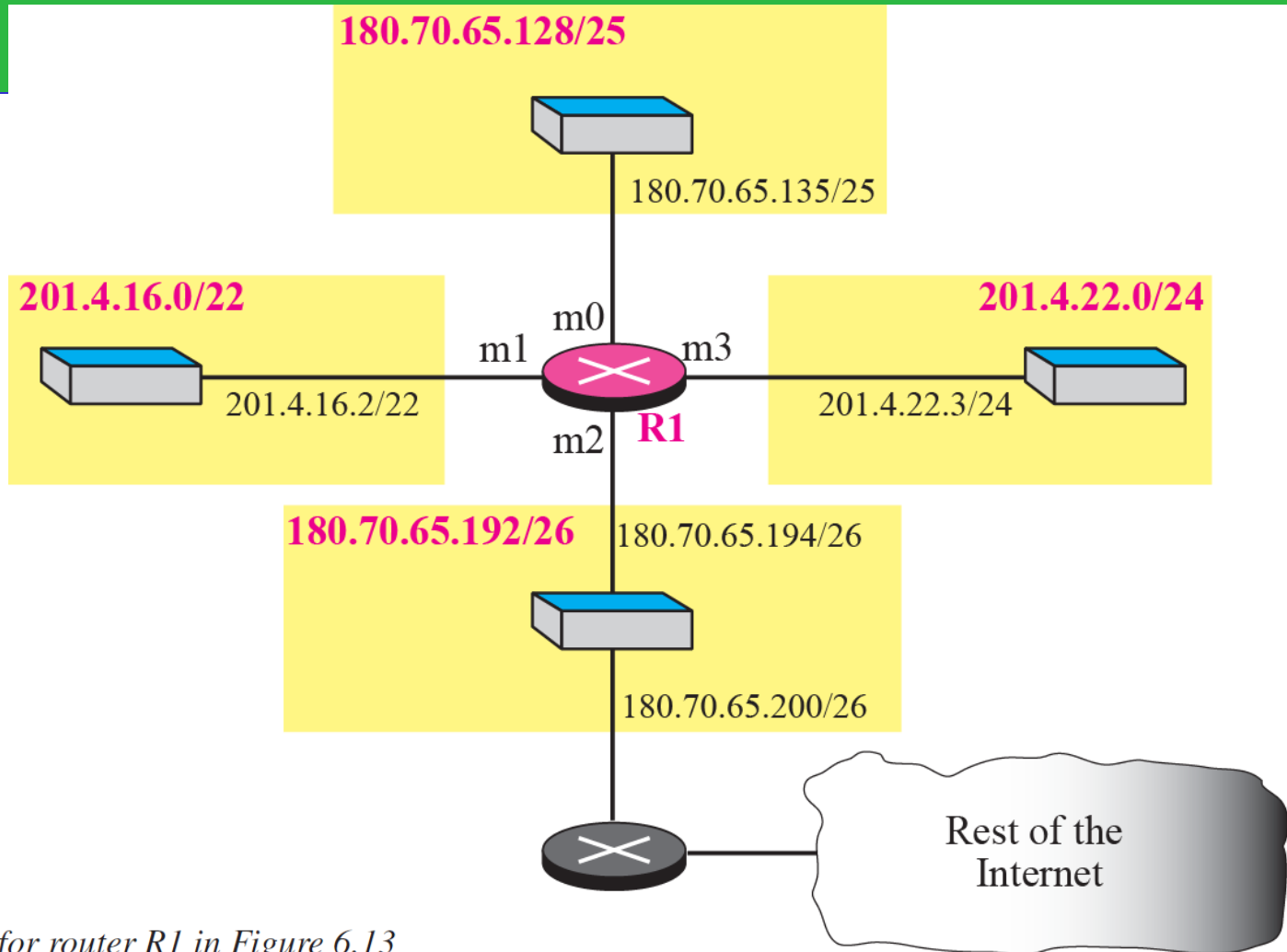


Table 6.1 Routing table for router R1 in Figure 6.13

Mask	Network Address	Next Hop	Interface
/26	180.70.65.192	-	m2
/25	180.70.65.128	-	m0
/24	201.4.22.0	-	m3
/22	201.4.16.0		m1
Default	Default	180.70.65.200	m2

Example

Below is the forwarding algorithm if a packet arrives at R1 with the destination address 180.70.65.140.

Routing steps:

The router performs the following steps:

- 1.** The first mask (/26) is applied to the destination address. The result is 180.70.65.128, which does not match the corresponding network address.
- 2.** The second mask (/25) is applied to the destination address. The result is 180.70.65.128, which matches the corresponding network address. The next-hop address (the destination address of the packet in this case) and the interface number m0 are passed to ARP

Example

Now let us give a different type of example. Can we find the configuration of a router if we know only its routing table? The routing table for router R1 is given in Table below. Can we draw its topology?

Table 6.2 *Routing table for Example 6.11*

<i>Mask</i>	<i>Network Address</i>	<i>Next-Hop Address</i>	<i>Interface Number</i>
/26	140.6.12.64	180.14.2.5	m2
/24	130.4.8.0	190.17.6.2	m1
/16	110.70.0.0	-----	m0
/16	180.14.0.0	-----	m2
/16	190.17.0.0	-----	m1
Default	Default	110.70.4.6	m0

Example

We know some facts but we don't have all for a definite topology.

- router R1 has three interfaces: m0, m1, and m2.
- there are three networks directly connected to router R1.
- there are two networks indirectly connected to R1.
- There must be at least three other routers involved (see next-hop column).
- not known if network 140.6.12.64 is connected to router R3 directly or through a point-to-point network (WAN) and another router.

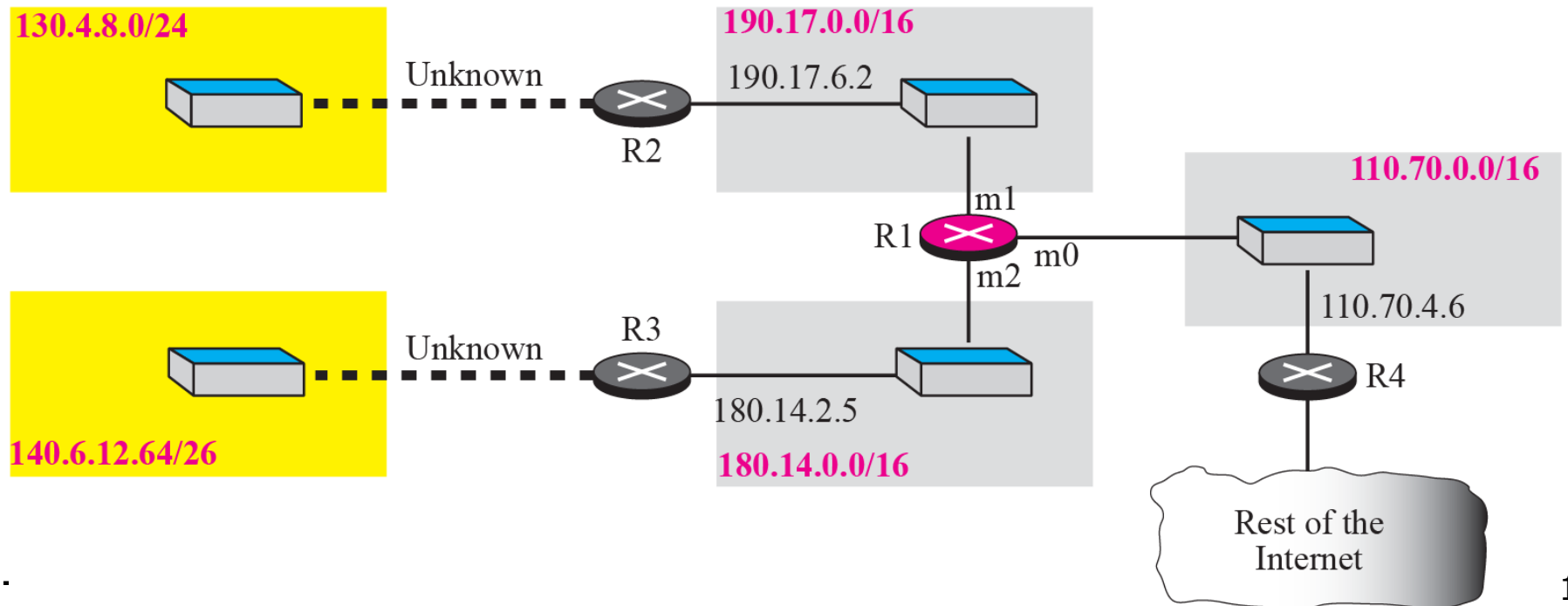
Table 6.2 *Routing table for Example 6.11*

<i>Mask</i>	<i>Network Address</i>	<i>Next-Hop Address</i>	<i>Interface Number</i>
/26	140.6.12.64	180.14.2.5	m2
/24	130.4.8.0	190.17.6.2	m1
/16	110.70.0.0	-----	m0
/16	180.14.0.0	-----	m2
/16	190.17.0.0	-----	m1
Default	Default	110.70.4.6	m0

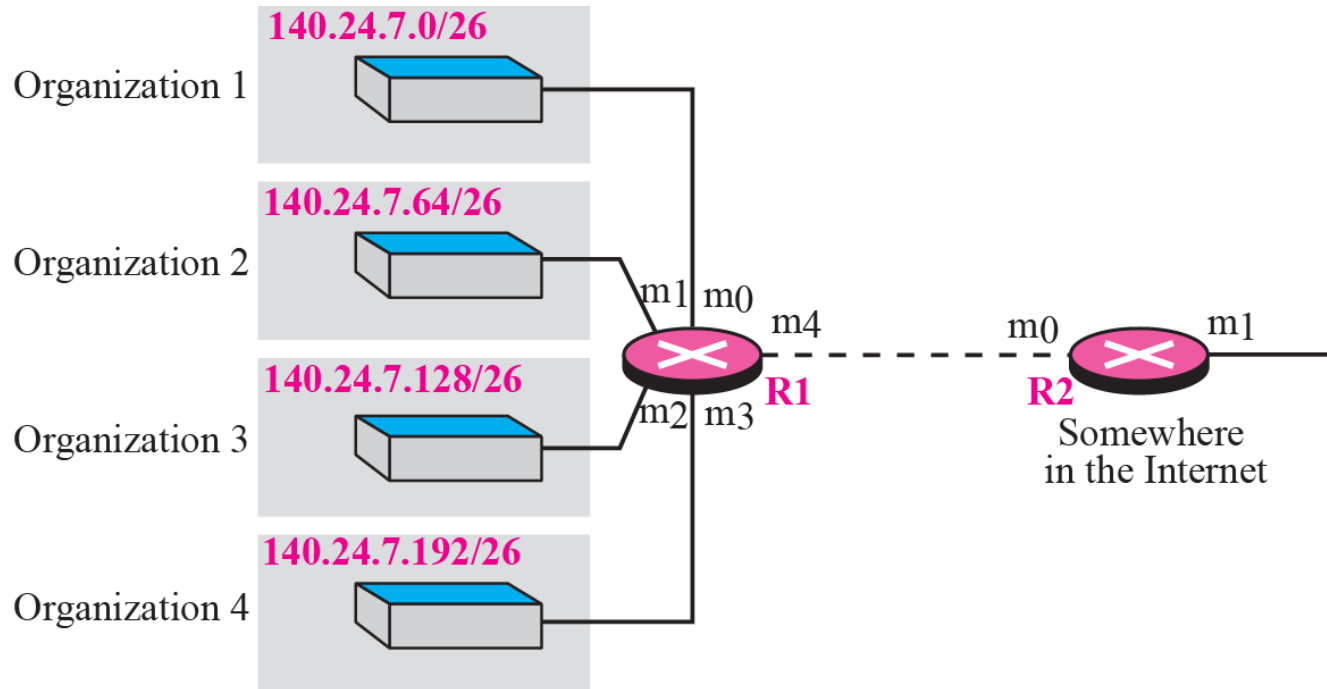
Example

We know some facts but we don't have all for a definite topology.

- router R1 has three interfaces: m0, m1, and m2.
- there are three networks directly connected to router R1.
- there are two networks indirectly connected to R1.
- There must be at least three other routers involved (see next-hop column).
- not known if network 140.6.12.64 is connected to router R3 directly or through a point-to-point network (WAN) and another router.



Address aggregation



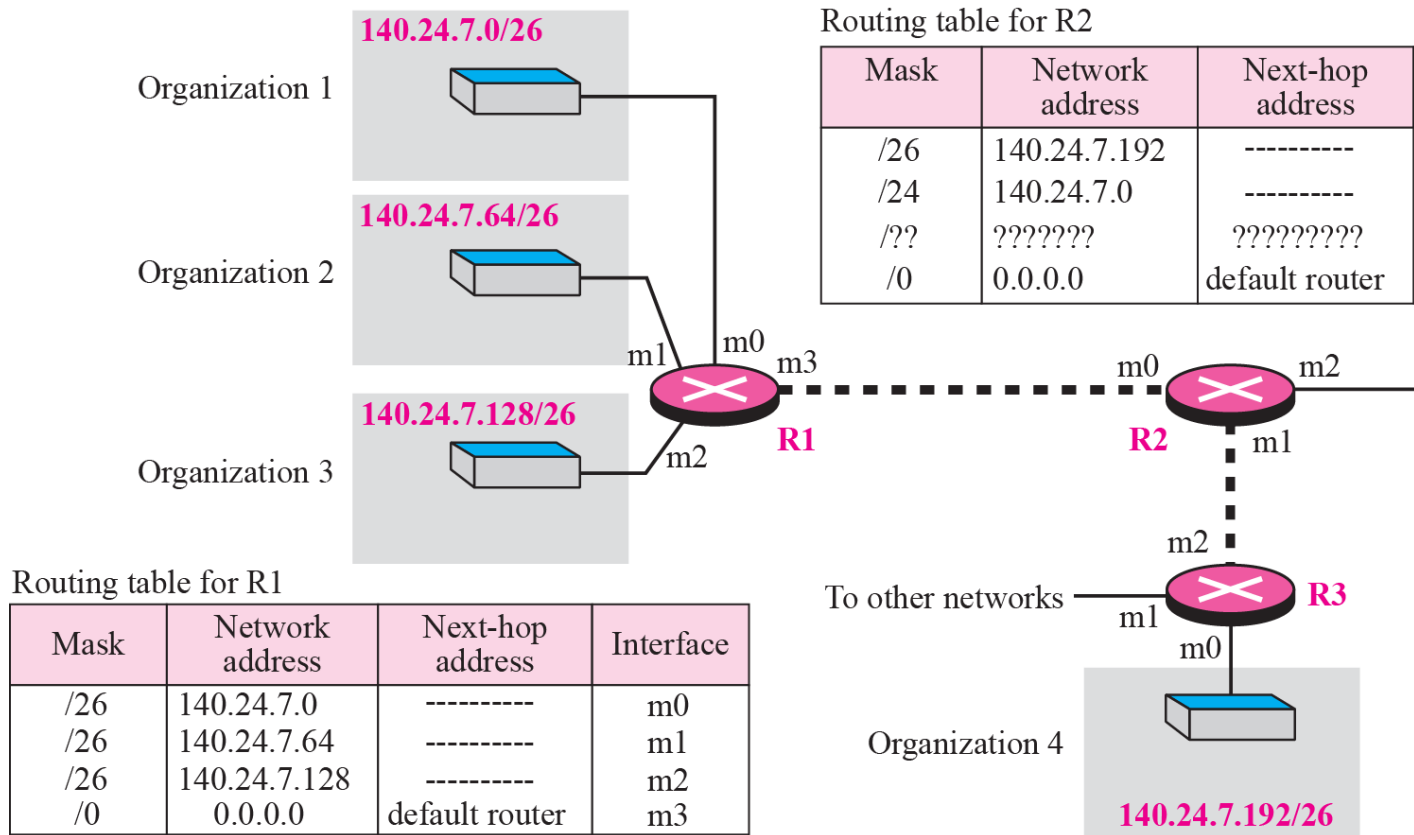
Mask	Network address	Next-hop address	Interface
/26	140.24.7.0	-----	m0
/26	140.24.7.64	-----	m1
/26	140.24.7.128	-----	m2
/26	140.24.7.192	-----	m3
/0	0.0.0.0	default router	m4

Routing table for R1

Mask	Network address	Next-hop address	Interface
/24	140.24.7.0	-----	m0
/0	0.0.0.0	default router	m1

Routing table for R2

Longest mask matching – 5 min exercise

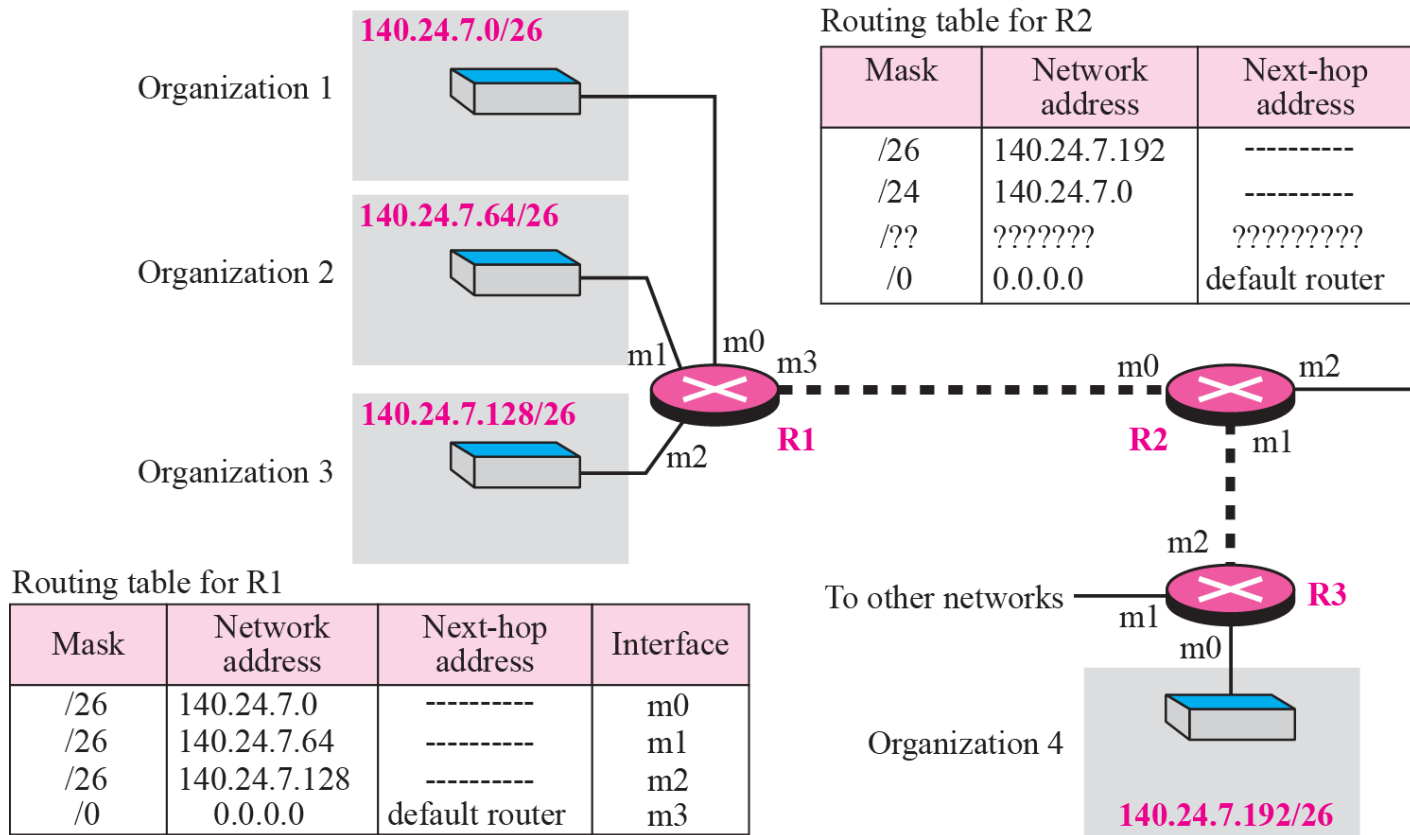


What is the matching interface for 140.24.7.232?

What is the matching interface for 140.24.7.104?

What is the algorithm for the longest match?

Longest mask matching – 5 min exercise



What is the matching interface for 140.24.7.232?

- m1

What is the matching interface for 140.24.7.104?

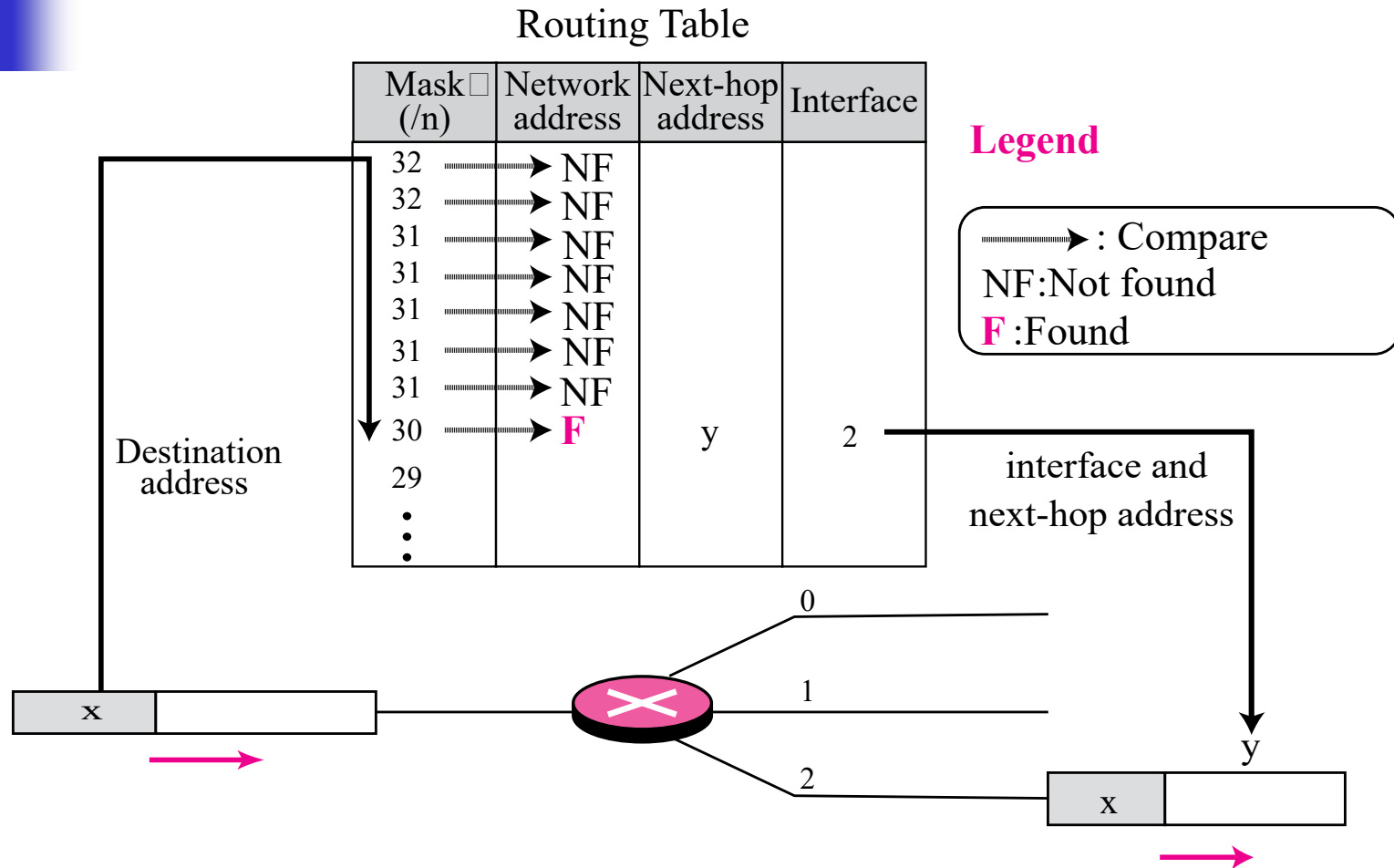
- m0

What is the algorithm for the longest match?

Routing table for R3

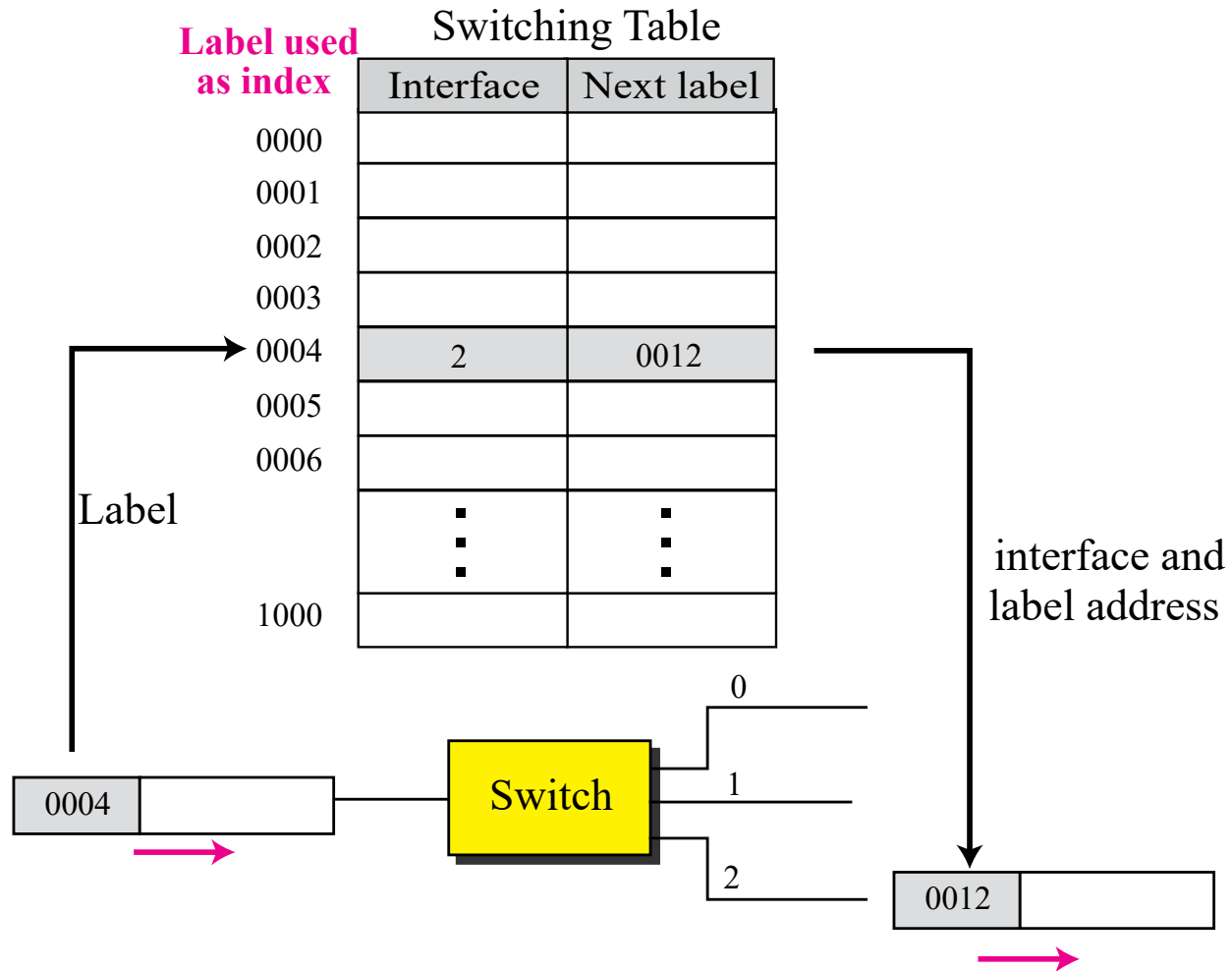
Mask	Network address	Next-hop address	Interface
/26	140.24.7.192	-----	m0
/??	???????	?????????	m1
/0	0.0.0.0	default router	m2

Algorithm for longest match based on destination address



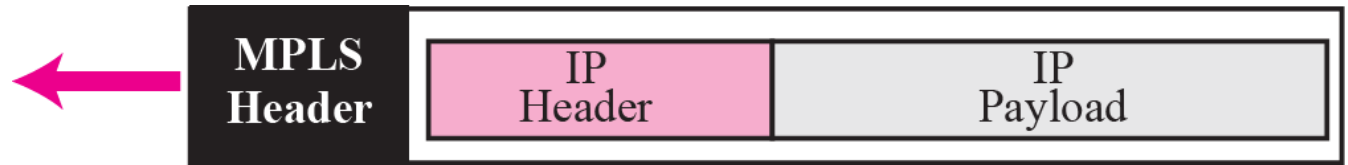
What is the challenge for the algorithm?

Access in table based on label

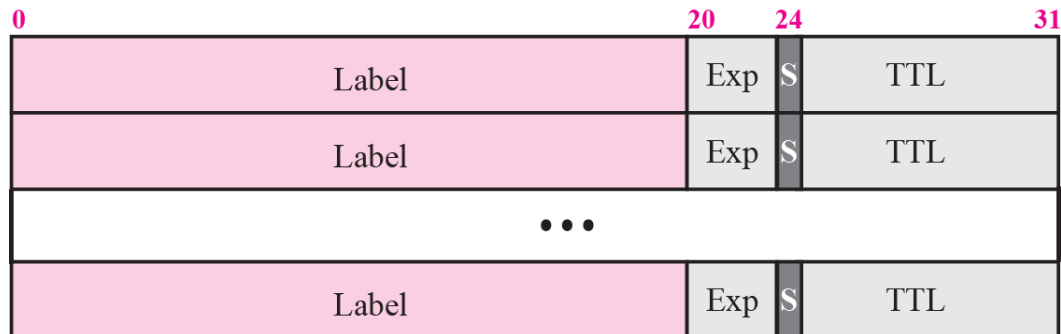


Labels are used as the index to the table, finding the information in the table is immediate.

MPLS header added to an IP packet

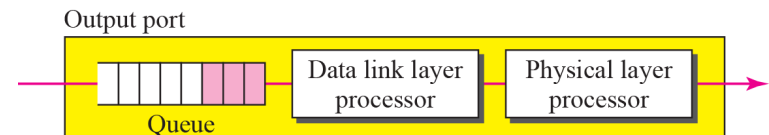
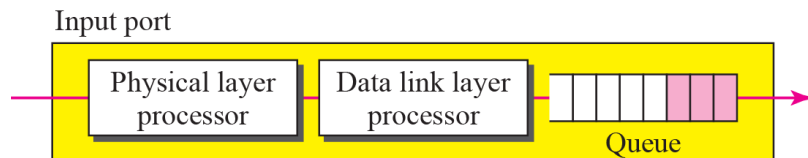
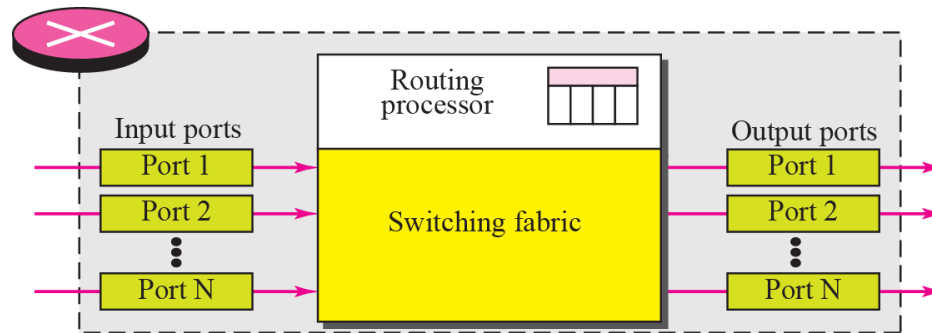


MPLS header made of stack of labels



STRUCTURE OF A ROUTER

- accepts incoming packets from one of the input ports (interfaces),
- uses a routing table to find the output port
- sends the packet from this output port.



References

**TCP/IP Protocol Suite (4th Edition),
by Behrouz A. Forouzan – Used the text's slide deck**

End of session